

Physiological Signal Variability for Pain Classification Using Random Forests

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Abstract—This project investigated whether variability in physiological signals could help distinguish Pain from No Pain using statistical feature engineering and machine learning. The analysis focused on diastolic blood pressure (BP Dia_mmHg), left-arm systolic blood pressure (LA Systolic BP_mmHg), electrodermal activity (EDA), and respiration rate. The dataset was cleaned by removing clearly failed sensor readings, including zero and negative values when they were physiologically invalid. For each signal, summary features including mean, standard deviation, maximum, minimum, variance, and median were generated. Time-series plots and box plots were used to compare variability across conditions, and separate Random Forest classifiers were trained for each physiological signal. The strongest individual test result was obtained from BP Dia_mmHg at approximately 73.3% accuracy, while LA Systolic BP_mmHg achieved the highest mean cross-validation accuracy at approximately 71.7%. Feature-importance analysis showed that standard deviation and variance were especially important for the blood-pressure signals. The results indicate that variability in physiological measurements, particularly blood pressure, contains useful information for pain classification while also showing the importance of careful preprocessing and multimodal interpretation.

1. Introduction

Pain is subjective and is typically assessed through self-report. Although self-report is clinically important, it may be unavailable or unreliable in situations where a patient cannot communicate effectively. For this reason, researchers have studied whether physiological measurements can provide objective evidence that complements traditional pain assessment. Physiological responses are promising because painful stimuli can influence autonomic activity, including cardiovascular responses, respiration, and skin conductance [1].

Machine-learning methods provide a way to convert physiological recordings into quantitative features and evaluate whether those features can distinguish pain-related states. Prior work has demonstrated that physiological features can support automatic pain detection [1], while multimodal research has shown benefits from combining physiological measurements with facial action information [2]. These findings motivate the study of individual physiological signals as interpretable building blocks for larger pain-recognition systems.

The objective of this DREAM research project was to determine whether statistical variability in selected physiological signals could help distinguish Pain from No Pain. The project examined diastolic blood pressure, left-arm systolic blood pressure, EDA, and respiration rate. The work included data cleaning, descriptive feature extraction, visualization, Random Forest classification, cross-validation, and feature-importance analysis.

2. Related Work

Automatic pain recognition has been studied using physiological, behavioral, and multimodal data. Winslow et al. [1] developed machine-learning models for pain detection using physiological features derived from heart-rate variability and respiration. Their results support the broader idea that pain-related physiological changes can be modeled computationally, while also illustrating the importance of selecting meaningful features from noisy biosignals.

The BioVid Heat Pain Database was introduced to support systematic evaluation of automated pain-recognition methods using video and biomedical signals [3]. Public datasets such as BioVid have helped establish pain recognition as a machine-learning problem in which sensor quality, feature design, and evaluation methodology strongly influence model performance.

Multimodal work is especially relevant to the present study. Hinduja, Canavan, and Kaur [2] investigated pain recognition by fusing physiological signals, including heart rate, respiration, blood pressure, and electrodermal activity, with facial action units. Their results showed that combining modalities can improve recognition compared with using only physiological signals or facial action units alone. That work used BP4D+, a multimodal spontaneous emotion corpus containing synchronized facial and physiological measurements, including blood pressure, respiration, skin conductance, and heart rate [4].

The current project takes a more focused approach by examining individual physiological signals and interpretable summary statistics. This provides a clear baseline for understanding which signals and which types of variability contribute useful information before moving to more complex multimodal fusion.

3. Methodology

3.1 Data Preparation

The analysis began with metadata containing 480 records and a signal matrix with 480 rows and 63,440 signal columns. After preprocessing and feature extraction, the machine-learning feature table contained 479 usable records and 10 columns, including subject information, signal type, condition, and the six engineered statistical features. Physiological recordings contained values that were clearly inconsistent with valid sensor measurements. Zero and negative values were removed when they represented failed or physiologically implausible readings, including unrealistic blood-pressure values. This step was necessary because invalid values can distort summary statistics, expand the apparent range of a signal, and influence a classifier in misleading ways.

3.2 Physiological Signals and Feature Extraction

Four physiological signal types were analyzed: BP Dia_mmHg, LA Systolic BP_mmHg, EDA, and respiration rate. For each signal, six descriptive features were generated: mean, standard deviation, maximum, minimum, variance, and median. Mean and median describe the central level of the signal, while standard deviation and variance capture dispersion. Minimum and maximum represent the observed range and can reveal unusually large changes during a condition.

3.3 Exploratory Analysis

Time-series plots and box plots were used to compare Pain and No Pain conditions. The visual analysis focused on differences in range and spread across conditions. This exploratory stage was important for identifying whether the engineered features were consistent with visible patterns in the physiological data before training machine-learning models.

3.4 Random Forest Classification

Separate Random Forest classifiers were trained for each physiological signal. The models used the statistical features for that signal to classify Pain versus No Pain. Random Forest was useful for this project because it can represent nonlinear relationships and also provides feature-importance estimates that help interpret which statistical descriptors contributed most to the prediction. Performance was evaluated using an individual test result as well as cross-validation mean accuracy.

4. Results

The visual analysis showed substantial variability in the blood-pressure signals. LA Systolic BP_mmHg stood out as having one of the widest ranges, especially during the Pain condition. This observation suggested that measures of dispersion could contain useful information for distinguishing Pain from No Pain.

The machine-learning results supported that observation. BP Dia_mmHg produced the strongest individual test accuracy at 73.3%, with a cross-validation mean accuracy of 66.7% (standard deviation 8.3 percentage points). LA Systolic BP_mmHg achieved 63.3% test accuracy but the highest cross-validation mean accuracy of 71.7% (standard deviation 7.2 percentage points). EDA achieved 53.3% test accuracy and 56.7% cross-validation mean accuracy, while respiration rate achieved 43.3% test accuracy and approximately 53.0% cross-validation mean accuracy. These results show that the blood-pressure signals were more informative individually than EDA or respiration in this analysis.

Signal	Test Acc.	CV Mean	CV Std.	Top Feature	Std + Variance Importance
BP Dia_mmHg	73.3%	66.7%	8.3%	Std	43.7%
EDA_microsiemens	53.3%	56.7%	5.7%	Variance	35.5%
LA Systolic BP_mmHg	63.3%	71.7%	7.2%	Std	48.2%
Respiration Rate_BPM	43.3%	53.0%	6.9%	Variance	35.4%

Table 1. Random Forest performance and variability-feature importance by physiological signal.

For BP Dia_mmHg, the confusion matrix was [[13, 2], [6, 9]], meaning 13 of 15 No Pain samples and 9 of 15 Pain samples in the test set were classified correctly. The model's Pain-class precision was 0.82 and recall was 0.60, while

the No Pain class had precision 0.68 and recall 0.87. This indicates that the model was relatively precise when labeling a sample as Pain, but it still missed a meaningful number of Pain cases.

For LA Systolic BP_mmHg, the confusion matrix was [[10, 5], [6, 9]]. Precision and recall were 0.64 and 0.60 for Pain and 0.62 and 0.67 for No Pain. EDA and respiration were weaker: EDA correctly classified 11 of 15 No Pain samples but only 5 of 15 Pain samples, while respiration correctly classified 8 of 15 No Pain samples and 5 of 15 Pain samples. These class-level results reinforce that the weaker signals did not separate Pain from No Pain as reliably when used alone.

Signal	Condition	Mean	Std	Variance
BP Dia_mmHg	No Pain	72.33	3.79	18.43
BP Dia_mmHg	Pain	77.94	6.99	86.13
EDA_microsiemens	No Pain	7.82	0.38	0.27
EDA_microsiemens	Pain	7.54	0.89	2.36
LA Systolic BP_mmHg	No Pain	118.78	3.46	15.38
LA Systolic BP_mmHg	Pain	115.93	9.15	165.78
Respiration Rate_BPM	No Pain	14.24	4.14	19.04
Respiration Rate_BPM	Pain	14.68	3.67	14.86

Table 2. Selected average feature values by signal type and condition.

The condition averages show why variability features became prominent. For BP Dia_mmHg, average standard deviation increased from approximately 3.79 in No Pain to 6.99 in Pain, while average variance increased from 18.43 to 86.13. The contrast was even larger for LA Systolic BP_mmHg: average standard deviation increased from 3.46 to 9.15 and average variance increased from 15.38 to 165.78 during Pain. EDA also showed higher variability during Pain, although its classification accuracy remained modest. Respiration did not show the same pattern; its average standard deviation and variance were slightly lower in Pain than in No Pain.

Feature-importance analysis provided an additional connection between the exploratory and machine-learning results. For BP Dia_mmHg, standard deviation was the single most important feature (importance 0.2248), followed closely by variance (0.2123). Together, these two variability features accounted for approximately 43.7% of total model importance. For LA Systolic BP_mmHg, standard deviation again ranked first (0.2457) and variance ranked second (0.2364), accounting for approximately 48.2% of total importance. EDA and respiration also assigned substantial weight to variability, with standard deviation plus variance contributing approximately 35.5% and 35.4% of total importance, respectively. These values provide direct quantitative support for the project's central finding that signal spread and fluctuation carry useful information for Pain versus No Pain classification, especially in the blood-pressure signals.

EDA and respiration were included in the same cleaning, feature-extraction, visualization, and Random Forest workflow. However, the completed progress report did not identify either signal as exceeding the strongest blood-pressure results. For that reason, the final interpretation emphasizes blood-pressure variability without claiming that a single signal provides a complete representation of pain.

5. Discussion

The results suggest that statistical variability in physiological signals can contribute to automated pain recognition. The strongest evidence in this project came from blood pressure, where both the classification results and feature-importance analysis pointed toward dispersion-based information. In particular, the importance of standard deviation and variance was consistent with the visual observation that blood-pressure signals exhibited substantial spread.

The difference between the best individual test result and the best cross-validation mean is also meaningful. BP Dia_mmHg produced the highest reported test accuracy, while LA Systolic BP_mmHg produced the highest reported cross-validation mean. This distinction illustrates why more than one evaluation view is useful: a single test split reflects one partition of the data, whereas cross-validation gives a broader view of performance across multiple partitions.

A major challenge was data quality. Physiological measurements can contain failed readings and can also be influenced by factors unrelated to pain. The project specifically identified stress, movement, breathing, and sensor quality as possible sources of variation. These factors make pain classification more difficult because a model may respond to general physiological arousal or artifacts rather than pain itself. Careful preprocessing and cautious interpretation are therefore essential.

The results also support the value of multimodal analysis. Prior work has shown that physiological signals and facial action information can provide complementary information [2]. The present project establishes an interpretable signal-

level baseline that could be extended by combining multiple physiological features and, where available, synchronized facial information.

6. Limitations and Future Work

Several limitations should be considered. First, physiological responses are not specific to pain and can change because of stress, movement, respiration, emotion, and sensor quality. Second, the project evaluated Random Forest models separately for each physiological signal rather than building a single multimodal model. Third, the final progress report provides the key accuracy and feature-importance outcomes but does not document every experimental setting needed to make broader claims about clinical generalization.

Future work could combine the physiological signals into a multimodal classifier and compare the combined model with the signal-specific baselines reported here. Additional work could also evaluate other model families, examine temporal or frequency-domain features, and investigate subject-independent validation. If synchronized facial data are available, facial action units could be incorporated to test whether multimodal fusion improves performance in the same way reported in prior work [2].

7. Conclusion

This project examined whether variability in physiological signals can help distinguish Pain from No Pain. After cleaning invalid sensor readings, the analysis generated mean, standard deviation, maximum, minimum, variance, and median features from diastolic blood pressure, left-arm systolic blood pressure, EDA, and respiration. Time-series plots and box plots were used to compare variability, and separate Random Forest classifiers were trained for each signal.

The strongest individual test result was approximately 73.3% for BP Dia_mmHg, while LA Systolic BP_mmHg achieved the highest mean cross-validation accuracy at approximately 71.7%. Standard deviation and variance were especially important for the blood-pressure models. Taken together, these results show that physiological variability—particularly blood-pressure variability—contains useful information for pain classification, while also emphasizing the importance of data quality, robust evaluation, and multimodal interpretation.

Acknowledgment

I thank Dr. Shaun Canavan for his guidance, feedback, and support throughout the DREAM research project. I also thank the DREAM Program for the opportunity to gain hands-on experience with data preparation, visualization, literature review, machine-learning development, evaluation, and technical research writing.

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